



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 • STANDARD 6.NOS.1

Division Expressions with Fractions and Mixed Numbers

Lesson 6-2 • Teacher Copy

Name: Type your name

Date: Today's date

Class: Class period



TODAY'S OBJECTIVES

Content Objective: I can divide a fraction by a fraction and divide with mixed numbers by multiplying by the reciprocal.

Language Objective: I can explain the steps using the words reciprocal, improper fraction, mixed number, and simplify.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Divide Fractions

Dividend

Divisor

Reciprocal

Quotient

Simplify

Mixed number

Improper fraction



Tap any word to see what it means and a picture.

1

Keep the first fraction, change $\div$ to $\times$, and flip the .

2

In $3/4 \div 1/2$, the fraction $3/4$ being divided is the .

3

In $3/4 \div 1/2$, the fraction $1/2$ we divide by is the .

4

To divide by $1/2$, I multiply by its , which is 2.

5

The answer to a division problem is the .

6

Writing a fraction in its smallest equal form, like $4/8 = 1/2$, is to

it.

7 A whole number and a fraction written together is a

 .

8 A fraction whose numerator is at least its denominator is an

 .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How does dividing fractions help the carpenter?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Divide Fractions** — Multiply the first fraction by the reciprocal of the second fraction.
Dividir fracciones — Multiplicar la primera fracción por el recíproco de la segunda fracción.

☐ **Dividend** — The number you are splitting up in a division problem.
Dividendo — El número que estás repartiendo en una división.

☐ **Divisor** — The number you split by in a division problem.
Divisor — El número entre el que divides en una división.

☐ **Reciprocal** — A fraction turned upside down.
Recíproco — Una fracción volteada de arriba abajo.

☐ **Quotient** — The answer when you divide.
Cociente — La respuesta cuando divides.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

The carpenter can cut ____ pieces because $5/6 \div 1/12 =$ ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: She will get many portions because $\frac{1}{8}$ pound is much smaller than $\frac{3}{4}$ pound.

Evidence: Students recognize the divisor $\frac{1}{8}$ is smaller than the dividend $\frac{3}{4}$, so the quotient is greater than 1.

Reasoning: This evidence connects the definition of mixed number to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The carpenter can cut ____ pieces because $\frac{5}{6} \div \frac{1}{12} =$ ____.
- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
- Mi afirmación es ____.*
- I know because ____.

Lo sé porque ____.

- So ____.

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**

Mi afirmación es ____.

- **My math evidence is ____.**

Mi evidencia matemática es ____.

- **This proves ____ because ____.**

Esto demuestra ____ porque ____.

4. Check Your Explanation

☐

I answered the question.

Respondí la pregunta.

☐

I used math evidence: numbers, an equation, a model, or a comparison.

Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

☐

I explained how the evidence proves my answer.

Explicué cómo la evidencia demuestra mi respuesta.

☐

I used at least two math words.

Usé por lo menos dos palabras matemáticas.

☐

I reread complete sentences and punctuation.

Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Divide Fractions
2. **Notes 2:** Dividend
3. **Notes 3:** Divisor
4. **Notes 4:** Reciprocal
5. **Notes 5:** Quotient
6. **Notes 6:** Simplify
7. **Notes 7:** Mixed number
8. **Notes 8:** Improper fraction
9. **Watch for this mistake:** *A common mistake in Divide Fractions is flipping the wrong fraction: students flip the dividend (the first fraction) instead of the divisor (the second fraction). For example, solving $3/4 \div 5/8$ by flipping $3/4$ to $4/3$ and keeping $5/8$ gives $4/3 \times 5/8 = 20/24 = 5/6$ — but only the divisor gets flipped. Keep $3/4$, change $\div$ to $\times$, and flip only the divisor $5/8$ to its reciprocal $8/5$: $3/4 \times 8/5 = 24/20 = 6/5$. Before you submit, check that the fraction you kept (the dividend) is still facing the same way it started.*
10. **Try It 1:** A. $4 - (2/3) \div (1/6) = (2/3) \times (6/1) = 12/3 = 4$. The dial unlocks at 4.
11. **Try It 2:** A. $3/2 - (3/4) \div (1/2) = (3/4) \times (2/1) = 6/4 = 3/2$. The keypad accepts $3/2$.